

Easy (10 MCQs)

1.

What is virtual networking?

- a) Physical cabling of network devices
- b) Creating logical networks over physical infrastructure
- c) Using wireless routers only
- d) None of the above

Answer: b

Explanation: Virtual networking creates logical (virtual) networks that run on top of physical hardware, allowing multiple networks to share the same physical infrastructure.

2.

Which technology is commonly used to segment networks virtually?

- a) VLAN
- b) HTTP
- c) FTP
- d) SMTP

Answer: a

Explanation: VLAN (Virtual LAN) allows logical segmentation of a physical network into separate broadcast domains.

3.

What does NAC stand for?

- a) Network Access Control
- b) Network Access Card
- c) Network Address Configuration
- d) None of the above

Answer: a

Explanation: NAC stands for Network Access Control, which manages device access policies on a network.

4.

Virtual Customer Edge primarily refers to:

- a) Physical customer routers
- b) Virtualized customer routing functions at provider edge
- c) Wireless network devices
- d) VPN endpoints

Answer: b

Explanation: VCE virtualizes customer edge functions at the provider's network edge, enabling more flexible management.

5.

Which protocol is commonly used for virtual overlay networks in datacenters?

- a) VXLAN
- b) FTP
- c) DNS
- d) SMTP

Answer: a

Explanation: VXLAN is used to create Layer 2 overlay networks over Layer 3 infrastructure in datacenters.

6.

Virtual networking helps improve:

- a) Security
- b) Scalability
- c) Network isolation
- d) All of the above

Answer: d

Explanation: Virtual networking improves security, scalability, and network isolation simultaneously.

7.

In virtual networking, what is the role of a virtual switch?

- a) To physically connect devices
- b) To route IP packets between networks
- c) To connect virtual machines within the same host or network
- d) None of the above

Answer: c

Explanation: Virtual switches operate like physical switches but connect virtual machines in virtual environments.

8.

Datacenter optimization through virtual networking helps primarily with:

- a) Faster internet browsing
- b) Resource utilization and automation
- c) Increasing cable length
- d) Reducing wireless interference

Answer: b

Explanation: Virtual networking helps optimize datacenters by improving resource utilization and enabling automation.

9.

Which of these is a benefit of Network Access Control?

- a) Allows unlimited network access
- b) Prevents unauthorized devices from connecting
- c) Increases network latency
- d) Slows down routing

Answer: b

Explanation: NAC enforces policies to restrict unauthorized device access to networks.

10.

Virtual networking abstracts:

- a) Physical topology
- b) Logical topology
- c) Physical media

d) None of the above

Answer: a

Explanation: Virtual networking abstracts the physical network topology to create logical networks.

Medium (30 MCQs)

11.

VXLAN operates at which OSI layer?

- a) Layer 1
- b) Layer 2
- c) Layer 3
- d) Layer 4

Answer: b

Explanation: VXLAN encapsulates Layer 2 Ethernet frames within Layer 3 UDP packets, effectively operating as a Layer 2 overlay.

12.

Which feature of virtual networking provides multi-tenancy in cloud environments?

- a) VLAN tagging
- b) IP routing
- c) MAC filtering
- d) DNS

Answer: a

Explanation: VLAN tagging enables multiple isolated tenant networks on shared physical infrastructure.

13.

In a virtualized environment, which component typically enforces network security policies?

- a) Physical firewall only
- b) Virtual firewall or security groups
- c) DNS server
- d) DHCP server

Answer: b

Explanation: Virtual firewalls or security groups enforce policies within virtual networks.

14.

Which technology allows creating virtual point-to-point WAN links over a shared physical network?

- a) GRE tunnels
- b) DHCP
- c) FTP
- d) VLAN

Answer: a

Explanation: GRE (Generic Routing Encapsulation) tunnels encapsulate packets to create virtual links.

15.

What is the main challenge when implementing virtual customer edge (VCE)?

- a) Increasing physical cabling
- b) Complexity in orchestration and management
- c) Wireless interference
- d) Slow routing speeds

Answer: b

Explanation: VCE requires sophisticated orchestration to manage virtualized routing functions efficiently.

16.

Which protocol is widely used to automate configuration in virtualized datacenters?

- a) DHCP
- b) NETCONF
- c) HTTP

d) FTP

Answer: b

Explanation: NETCONF is used for programmatic configuration of network devices.

17.

In Network Access Control, what happens to a non-compliant device?

- a) It is granted full network access
- b) It is quarantined or denied access
- c) It is given administrative privileges
- d) It is automatically upgraded

Answer: b

Explanation: NAC quarantines or denies access to devices that don't meet security policies.

18.

Which of the following is NOT a benefit of virtual networking?

- a) Improved flexibility
- b) Network isolation
- c) Physical device replacement only
- d) Centralized management

Answer: c

Explanation: Virtual networking adds flexibility and isolation but does not merely replace physical devices.

19.

What does VLAN trunking allow?

- a) Transmit multiple VLANs over a single physical link
- b) Create physical networks only
- c) Disable routing between VLANs
- d) None of the above

Answer: a

Explanation: VLAN trunking tags multiple VLANs on one physical link to carry traffic for multiple VLANs.

20.

What is the maximum VLAN ID value in the IEEE 802.1Q standard?

- a) 100
- b) 4094
- c) 1024
- d) 2048

Answer: b

Explanation: VLAN IDs range from 1 to 4094 as per 802.1Q.

21.

VXLAN uses which UDP port by default?

- a) 4789
- b) 80
- c) 443
- d) 67

Answer: a

Explanation: VXLAN uses UDP port 4789 for encapsulated traffic.

22.

Which of the following best describes a virtual switch?

- a) A software-based switch in virtual environments
- b) A hardware switch
- c) A wireless access point
- d) None of the above

Answer: a

Explanation: A virtual switch is a software switch used to connect virtual machines.

23.

In virtual networking, isolation between tenants is mainly provided by:

- a) IP addressing only
- b) VLANs or overlays
- c) Physical separation only
- d) DNS

Answer: b

Explanation: VLANs or overlay technologies like VXLAN provide tenant isolation.

24.

Which virtual networking concept allows dynamic assignment of policies based on user roles?

- a) Network Access Control (NAC)
- b) Static routing
- c) MAC address filtering
- d) None of the above

Answer: a

Explanation: NAC enforces dynamic policies based on device and user compliance.

25.

What role does SDN play in virtual networking?

- a) Centralizes control plane for programmability
- b) Adds physical cabling
- c) Slows down routing
- d) None of the above

Answer: a

Explanation: SDN separates control and data planes, centralizing control for easier network management.

26.

Which encapsulation method is used by VXLAN?

- a) Ethernet in UDP
- b) IP in IP
- c) MPLS

d) None of the above

Answer: a

Explanation: VXLAN encapsulates Layer 2 Ethernet frames within UDP packets.

27.

Which protocol can be used to secure communication between virtual networking components?

- a) TLS
- b) FTP
- c) HTTP
- d) DNS

Answer: a

Explanation: TLS secures communication channels between network components.

28.

Which of these is a key consideration for datacenter virtual networking scalability?

- a) Number of physical cables
- b) Overlay network size and addressing
- c) Color of hardware devices
- d) None of the above

Answer: b

Explanation: Scalability depends on overlay network design and addressing schemes.

29.

Virtual networking contributes to disaster recovery by:

- a) Physical backup only
- b) Enabling rapid network provisioning on new hardware
- c) Ignoring backup procedures
- d) None of the above

Answer: b

Explanation: Virtual networking enables quick restoration by easily provisioning networks on alternate hardware.

30.

What is a major challenge with virtual networking security?

- a) Ensuring proper segmentation and policy enforcement
- b) Increasing cable length
- c) Reducing wireless interference
- d) None of the above

Answer: a

Explanation: Proper segmentation and enforcement of security policies in virtual environments is complex.

Hard (10 MCQs)

31.

In VXLAN, what component is responsible for mapping between VNIs and physical network?

- a) VTEP (VXLAN Tunnel Endpoint)
- b) DHCP server
- c) NAT device
- d) DNS server

Answer: a

Explanation: VTEP encapsulates and decapsulates VXLAN traffic and maps VNIs to physical networks.

32.

Which mechanism ensures tenant isolation in multi-tenant virtual networks over shared infrastructure?

- a) VLAN and VXLAN segmentation with access control lists (ACLs)
- b) Physical cabling separation only
- c) MAC address spoofing

d) None of the above

Answer: a

Explanation: VLAN/VXLAN along with ACLs enforce tenant isolation.

33.

In SDN-based virtual networking, the separation of control and data planes primarily enables:

- a) Centralized management and programmability
- b) Decentralized routing only
- c) Increased latency
- d) None of the above

Answer: a

Explanation: Separation allows centralized control and programmability of networks.

34.

Which of the following can be a bottleneck in virtualized datacenter networks?

- a) Physical NIC bandwidth and CPU resources on hypervisors
- b) Wireless interference
- c) Color of the server rack
- d) None of the above

Answer: a

Explanation: NIC bandwidth and hypervisor CPU impact network throughput in virtualized setups.

35.

Which layer of the TCP/IP stack does VXLAN encapsulation primarily operate on?

- a) Transport layer (UDP)
- b) Application layer
- c) Session layer
- d) None of the above

Answer: a

Explanation: VXLAN encapsulates Ethernet frames inside UDP packets at the transport layer.

36.

How does NAC typically determine device compliance before granting access?

- a) Checking device health status via posture assessment
- b) By physical inspection only
- c) By guessing
- d) None of the above

Answer: a

Explanation: NAC assesses devices' compliance with security policies before access.

37.

Which tool would you use to centrally orchestrate virtual network services in an SDN environment?

- a) SDN Controller
- b) DHCP Server
- c) DNS Server
- d) None of the above

Answer: a

Explanation: SDN Controllers manage and orchestrate network devices and services.

38.

How is scalability typically achieved in virtual networking for service providers?

- a) Using hierarchical overlays and automation
- b) Adding more physical switches only
- c) Manual reconfiguration only
- d) None of the above

Answer: a

Explanation: Hierarchical overlays and automation help scale virtual networks efficiently.

39.

Which feature allows dynamic isolation of devices in virtual networking based on user identity?

- a) Role-based access control (RBAC) integrated with NAC
- b) Static VLAN assignment only
- c) Static routing
- d) None of the above

Answer: a

Explanation: RBAC combined with NAC provides identity-based dynamic isolation.

40.

Which technology complements virtual networking by providing network function virtualization (NFV)?

- a) Virtualized firewalls, load balancers, and routers running as software
- b) Physical routers only
- c) Wireless access points only
- d) None of the above

Answer: a

Explanation: NFV runs network functions as software, complementing virtual networks.

41.

What is the impact of virtual networking on disaster recovery?

- a) Enables quick network restoration via automated provisioning
- b) Makes recovery slower
- c) Requires physical rewiring
- d) None of the above

Answer: a

Explanation: Virtual networking allows rapid re-provisioning, improving recovery times.

42.

Which of the following is an essential security consideration for virtual customer edge deployments?

- a) Secure authentication and encryption of control plane
- b) Ignoring control plane security
- c) Relying on physical security only

d) None of the above

Answer: a

Explanation: Securing control plane communication is critical for virtual edge devices.

43.

Which encapsulation protocol besides VXLAN is often used in virtual networking overlays?

- a) GRE
- b) FTP
- c) SMTP
- d) DNS

Answer: a

Explanation: GRE tunnels also provide overlay capabilities in virtual networks.

44.

How does virtual networking support cloud-native applications?

- a) By providing micro-segmentation and policy automation
- b) By requiring manual network setup only
- c) By blocking automation
- d) None of the above

Answer: a

Explanation: Virtual networks enable micro-segmentation and automation needed for cloud-native apps.

45.

Which protocol is used to carry virtual network identifiers (VNIs) in VXLAN?

- a) UDP
- b) TCP
- c) ICMP
- d) FTP

Answer: a

Explanation: VNIs are carried inside VXLAN encapsulated UDP packets.

46.

What is the role of access control lists (ACLs) in virtual networking?

- a) To filter and control traffic between virtual segments
- b) To increase packet size
- c) To slow network speeds
- d) None of the above

Answer: a

Explanation: ACLs enforce traffic filtering and security between virtual networks.

47.

In virtual networking, what does 'overlay' mean?

- a) A virtual network built on top of a physical underlay network
- b) A physical cable
- c) An internet browser extension
- d) None of the above

Answer: a

Explanation: Overlay networks use tunneling to create virtual networks over physical infrastructure.

48.

Which is a common challenge in managing virtual networks?

- a) Monitoring and troubleshooting due to abstraction layers
- b) Physical cable damage only
- c) Color mismatch in hardware
- d) None of the above

Answer: a

Explanation: Abstraction layers can complicate monitoring and troubleshooting.

49.

Which service in virtual networking helps maintain QoS (Quality of Service)?

- a) Traffic shaping and prioritization in virtual switches
- b) DNS service only
- c) FTP
- d) None of the above

Answer: a

Explanation: Virtual switches and controllers apply QoS policies to prioritize traffic.

50.

What is the benefit of using software-defined networking (SDN) in virtual networks?

- a) Centralized programmability and automation
- b) Physical cabling replacement only
- c) Manual configuration only
- d) None of the above

Answer: a

Explanation: SDN centralizes control, enabling automation and programmability.